Raspberry Pi vs ESP32-S3: why live video is harder on the ESP32

Notes from moving the ShelfEye shelf camera from a Raspberry Pi to an ESP32-S3 camera board.

Hardware compared

	Raspberry Pi (Zero W / Zero 2 W)	ESP32-S3 CAM board
Fast memory	512 MB	512 KB (about 1000x less), plus 8 MB slower PSRAM
Processor	Zero W: 1 core, 1 GHz. Zero 2 W: 4 cores, 1 GHz	2 cores, 240 MHz
Camera	Camera Module 3; the Pi has a hardware video encoder and enough power to make JPEGs quickly	GC2145 gives raw images: each 640x480 frame is about 600 KB, which the ESP32 converts to JPEG itself

What this means for ShelfEye

- **Counting works on both.** It needs one photo every 3 minutes, converted and sent once, which is light work even for the ESP32.
- **Live view is slow on the ESP32.** For every frame it must grab a raw image, convert it to JPEG and send it over an encrypted connection (about 40 KB of the fast memory) through Render, while Wi-Fi, the camera and counting keep running. Expect roughly 1-3 frames a second at 640x480 with the GC2145 (an estimate, not yet measured).
- **Smooth video (25-30 frames a second) from anywhere** needs Pi-class hardware and a video streaming service, which uses far more power than a battery-run ESP32.

The camera module

The board was sold as having an **OV3660**, but it reports a **GC2145** (chip ID 0x2145), and it can't produce JPEG images.

	OV3660 (advertised)	GC2145 (received)
Resolution	3 MP (2048x1536)	2 MP (1600x1200)
JPEG made by the camera	Yes	No; the ESP32 converts each image
Image quality	Better	Lower, especially in dim light
Live view speed on ESP32	Several times faster	Slow

An OV2640 or OV3660 module fits the same socket. The firmware detects the camera at start-up, so swapping modules needs no code changes.